

ControlPlane.ai — Round 2 Detailed Business Proposal

Accenture Innovation Challenge 2026 · Problem Track 1 · Team ControlPlane

Choda Srujan Sai · Dhrithika — IIT Gandhinagar

Companion to the frozen system of record: [ARCHITECTURE.md](#).

Positioning: [NARRATIVE.md](#). Hostile Q&A: [QA.md](#).

Prototype: `python3 examples/refund_trace_demo.py` and `python3 examples/multi_usecase_demo.py`.

This document is the Round 2 business proposal. It cites the architecture; it does not reopen it.

Business-proposal truth source: [round2/CONTROLPLANE_R2_FINAL.md](#). Branch of record: main.

Stage Check (compact)

Stage	Locked into	Status
R2S1 prototype scope	§§1–4	PASS
R2S2 enterprise envelope	§5	PASS
R2S3 build/demo	§6	PASS
R2S4 buyers/value/rollout	§§7–11	PASS

#	Eternal invariant	Status
1	Default = UNSUPPORTED	PASS
2	Entitlement = <code>span.acl</code> $\subseteq$ <code>principal.clearance</code> ; zero LLM	PASS
3	Exact $R \times S$ matrix; never redrawn; no route parameter	PASS
4	One graph: <code>STEP</code> $\rightarrow$ <code>SPAN</code> $\rightarrow$ <code>CLAIM</code> $\rightarrow$ <code>ACTION</code>	PASS
5	Hard gate on actions , not tokens	PASS
6	Dual-action: R1 Edit + R3 Escalate (held , never “blocked”)	PASS
7	UNKNOWN never $\rightarrow$ SUPPORTED	PASS
8	FNR as typed format; empty until earned	PASS
9	Bias = async route-level only (never live matrix cell)	PASS
10	Refuse-to-claim list (about us)	PASS

1. Problem framing

An AI response is not text to be scored — it is a set of **claims requesting permission to act**.

Enterprises no longer run one assistant. They run three at once: a customer-facing support chatbot, an internal knowledge copilot, and a decision-support agent that can refund, file, send, or write to production. Each has a different blast radius, a different latency budget, and a different owner. Oversight tooling is still built for the answering era: score the finished paragraph, chart the failure, review the log next week.

That category error now has a rupee figure attached.

A refund agent emits: *“Approved. Refund of ₹1,84,000 issued under clause 7.2 of the vendor agreement.”* Every filter passes it. Confidence 0.94. Money moves Tuesday, found Friday. **Clause 7.2 does not exist.** The company wrongly pays out ₹1,84,000. The system did not fail a classifier. **It was never asked to prove anything.**

The cost of a wrong output changed category: it used to be a bad paragraph. It is now an executed transaction.

Three things make this hard in a real enterprise, and they are the same three the Round 2 brief names:

1. **One-size-fits-all checking fails.** A hedged warranty guess on a support reply is not the same object as an ungrounded clause authorizing a payment. Treating them as one “risk score” either over-flags the first (alert fatigue, the plane gets switched off) or under-flags the second (liability).
2. **There is often no ground truth at decision time.** The knowledge gap that caused the hallucination is the same gap a checker would need. If the plane requires a labelled gold answer, it cannot stand in front of an action.
3. **The model is consumed via API.** There are no weights, no logits, no hidden states to inspect. Anything that needs model internals is a paper design.

ControlPlane is an **admission-control layer**. It captures the evidence the model was actually given *before generation*, binds each claim in the output back to a span of that evidence, and refuses to let an unproven claim cross into an action. Verification effort is priced by blast radius: a draft is checked cheaply, a payment is not.

We do not claim to eliminate hallucinations. We claim that ungrounded claims cannot authorise actions, and we report what we miss.

2. Solution design

2.1 One graph, three axes

Everything downstream reads a single structure, assembled *while* the response is produced rather than reconstructed afterwards ([ARCHITECTURE.md §2](#)):

STEP	produces	SPAN	binds	CLAIM	authorizes	ACTION
(tool call, retrieval, model turn)	(retrieved chunk, tool row, DB record – with source_id, ACL, hash, offsets)	(typed atomic proposition from the output stream)	(pending side effect: tool + args + irreversibility)			

Performance reads it forward — does each claim bind to a span?

Cost reads it backward — did each step produce a span that grounded any accepted claim?

Responsibility reads its labels — is the caller entitled to every span a claim binds to, and does the action fall inside its typed interlock?

One structure, three axes, three actuators. That is why this is a control plane and not three classifiers in a trench coat.

The verdict is a **set-membership test** against receipts captured outside the model — not an opinion about finished text. Default verdict is **UNSUPPORTED**. A claim must earn **SUPPORTED**. **UNKNOWN** is a first-class

verdict that *routes*; it never collapses into SUPPORTED.

2.2 The keystone: Provenance Recorder

If exactly one thing gets built, build the Provenance Recorder ([ARCHITECTURE.md](#) §2, §12). It hooks **context assembly**, not the model. Every span is captured with `source_id`, ACL, content hash and offsets. Context then freezes: the model cannot invent a span after the fact, and it has no channel to declare a binding.

Every other mechanism degrades to a generic guardrail without this hook. Model-emitted citations are rejected for the same reason: a model that fabricates a fact fabricates the citation with equal fluency ([ARCHITECTURE.md](#) §8).

The Round 2 prototype *is* this keystone, plus the three reads that become possible once it exists: deterministic binding, entitlement as ACL set-membership, and the Action Interlock on the frozen blast-radius matrix. See [Appendix A](#).

2.3 Decision policy — transcribed, never redrawn

Two inputs govern every decision: **blast radius R** (what this response can break) and **verdict severity S** (what the evidence says).

R = irreversibility × audience × data class × autonomy level

Tier	Meaning
R0	internal draft, a human reads it first
R1	user-visible, read-only
R2	reversible write / external send
R3	irreversible or regulated — payment, deletion, publication, regulated advice

	Contradicted entitlement violation	Unsupported categorical	Unsupported hedged	Unknown
R3	Block	Escalate	Escalate	Escalate
R2	Block	Edit	Edit	Escalate
R1	Edit	Edit	Pass + annotate	Pass + annotate
R0	Pass + annotate	Pass + annotate	Pass	Pass

The verdict is hostile; the action is proportionate. A claim must earn SUPPORTED; failing to earn it maps to the frozen cell for that pending action — Pass / Pass + annotate / Edit / Escalate / Block — never a single response-level “blocked.” That is the over/under-flag tradeoff, encoded rather than “tuned away.”

Actuators are exactly **Block · Edit · Escalate · Pass / Pass + annotate** (plus autonomy downgrade and circuit breaker, spoken not drawn). Forbidden inventions: STREAM, Kill Span, Hold & Re-verify, Redact & Flag, COMMIT BLOCKED as the R3 unsupported-categorical label.

Edit is surgical, never generative. Strip the unsupported or unentitled claim. Free-form rewriting produces a new unverified artifact.

Escalate ships an evidence packet, not an alert: the claim, the candidate spans, the verdict, the diff. Dual-action centrepiece: R1 **Edit** + R3 **Escalate** — refund is **held**, never “blocked.”

bullet Fail stance belongs to the **tier**, not a global default: R0/R1 fail open with annotation; R2/R3 fail closed or escalate. A universal fail-open makes the plane bypassable by anyone who can induce load.

bullet **Absence of a decision is not a permission.** Two silent-pass paths were closed in the prototype: a binding that cites a span the ledger never recorded cannot earn SUPPORTED (it falls to UNSUPPORTED), and an action with **no routed claims** is priced through the frozen UNKNOWN column instead of passing by default. Entitlement is always computed by the plane; a caller-supplied finding may raise a verdict, never clear one. No new matrix cell was added to do this (Appendix A.4).

User surface: three states — **Verified / Uncertain / Blocked** — one claim-level line each. No raw scores. Refund action surface remains **Held / Escalate**, not “blocked.”

2.4 Latency and deployment shape

The expensive part is precomputed: provenance is indexed at context-assembly time, so binding is a lookup against a small in-memory set, not a search over a corpus.

bullet Hard gate sits on **actions**, not tokens. Users perceive speed from text; harm lives in the commit path.

bullet Three lanes. Lane 1 — inline, deterministic only (span membership, ACL, typed interlocks, arithmetic). Lane 2 — NLI binding for flagged claims and high-R traffic. Lane 3 — async (semantic-entropy, counterfactual bias replay, shadow audit).

bullet **Budget follows blast radius.** R0/R1 is 80–90% of volume and gets Lane 1 only.

bullet Targets: **≤40 ms p50 and ≤200 ms p95** added on R0/R1 text. Never quote 40 ms as p95.

bullet Deployment: an **OpenAI-compatible reverse proxy** plus a thin context-assembly SDK hook. No application rewrite, no model access, no weights, no logits, no fine-tuning. Integration is real and is the moat — we say so.

The prototype implements Lane 1 of this design on fixtures (no LLM, no network). Lane 2/3 and the proxy are roadmap, not pretended.

2.5 Why the overlapping failures are not a category crisis

The brief notes that bias, hallucination and privacy often overlap — a fabricated detail about a person can be both a hallucination and a privacy concern. We do not collapse them into one classifier. They have different mathematics, error costs and owners, and they land as **different labels on the same graph**:

Apparent mix	What the graph actually says	Who acts
Fabricated personal detail (no span)	Performance: UNSUPPORTED. Leakage: PII-shaped entity binding to no span = model-memory leak → Block	Entitlement Auditor + Interlock
True HR fact from an over-permissioned index	Performance: SUPPORTED (it is in the evidence). Responsibility: span ACL excludes the caller → entitlement violation → Block on R2/R3, Edit on R1	Entitlement Auditor — structurally impossible for any output-only checker
Decision-shaped output that flips when a protected attribute is perturbed	Not a per-response property. Bias = counterfactual invariance, route-level, async: decision flip rate with a confidence interval over a rolling window. Flag when the CI excludes zero.	Adjudicator (Lane 3)

Do not drop bias. State it in measurement terms, never moral ones.

2.6 What we refuse to claim

About ourselves, not competitors (NARRATIVE.md §5):

- bullet We do **not** eliminate hallucinations.
- bullet We do **not** claim zero integration or “drop it in.” We hook context assembly.
- bullet We do **not** claim zero added latency. We never make the model feel slow; we make the action wait.
- bullet We do **not** quote one accuracy number across performance, cost and responsibility.

3. Target users

One plane, three buyers. They share a ledger and disagree about which column they read.

User	Reads	Job on the plane	What they will switch it off for
Platform / SRE	Cost axis + latency SLOs + circuit breaker	Own the reverse proxy and the SDK hook. Kill runaway loops (non-convergence breaker). Keep p50/p95 inside budget. Canary and auto-rollback threshold changes.	Added latency on R0/R1 text; a plane that is a single point of failure for the whole product
Risk / compliance / audit	Responsibility axis + append-only ledger	Entitlement (who was allowed to know); typed interlocks on irreversible actions; hash-chained evidence packet on every hold. Policy as a versioned DAG, not a slide.	A composite 0–100 score they cannot decompose; a miss they cannot explain to a regulator
App owners — support, copilot, agents	Performance axis + the matrix cell for <i>their</i> route	Support owner wants replies to stream. Copilot owner wants drafts edited, not blocked. Agent owner wants payments held until a claim binds. Enforcement is earned per route through shadow evidence.	Over-blocking of ordinary text; a global switch that treats a chatbot like a wire

The same pending response can serve two of them at once. In the refund running example the customer-visible text is **Edited** (R1 × entitlement) while the refund is **held and escalated with the evidence packet** (R3 × unsupported-categorical). Both are correct simultaneously. Proof scales with consequence.

4. Business case & impact

Reference parameters from the Round 2 brief, adopted as directional assumptions (not a measured dataset):

Parameter	Directional assumption
Weekly volume	~ 40,000 interactions/week across the three use cases combined (“tens of thousands”)
Mix	Support chatbot ~70% (R1, latency-sensitive); internal copilot ~25% (R1/R2); decision-support / acting agents ~5% (R2/R3)
Blast-radius volume	R0/R1 $\approx$ 80–90% of traffic → Lane 1 only (ARCHITECTURE.md §5)
Data	Mix of well-governed sources (ACL'd systems of record) and loosely governed ones (wikis, FAQs, pasted tickets)
Unit of irreversible harm	The running example: ■1,84,000 wrongly paid on an ungrounded clause

We do **not** put a savings percentage, a catch rate, or a payback period on this page. Those numbers are knowable only on the buyer's traffic, which is why the gate report ships as an **empty schema with typed placeholders** ([ARCHITECTURE.md](#) §10.7). What is knowable at design time is *which quantities the graph makes exact*.

4.1 Three quantities a buyer can cash

Held irreversible actions. $R3 \times$ unsupported-categorical does not “block the refund” as a slogan — it **holds and escalates with the evidence packet**. On ~2,000 decision-support interactions/week, even a thin tail of R3 attempts is a budget line: a single miss at the running-example amount is already material, and the plane's job is to make that miss a held packet rather than a Tuesday payout found Friday. The prototype produces this actuator on the real matrix cell $R3 \times \text{Unsupported} + \text{categorical}$ (Appendix A).

Entitlement incidents prevented. The most common real enterprise incident is not a jailbreak. It is an over-permissioned RAG index faithfully leaking HR data to the wrong employee. Output-only wrappers cannot catch it: they never carry caller identity into verification. ControlPlane does — ACL set-membership, microseconds, near-zero FP. The prototype's `hr_side` claim is this incident: `span ACL {hr-confidential} vs caller clearance {vendor-public} → Edit` on the R1 text path.

Dead compute, named exactly. Walk the graph backward: every SUPPORTED claim in the accepted answer points to a span, which points to the step that produced it. Any step grounding zero accepted claims is dead compute — no model, no estimation. In the refund fixture, five assembly steps run; `faq_search` and `crm_lookup` ground nothing. A dashboard can say the trace cost **■8**. Walking the graph can say **■5** of it grounded nothing. We will measure that yield on their traffic in shadow; we will not invent the percentage here.

4.2 Why this is cheaper than the alternative at 40k/week

Uniform deep checking at this volume is how planes get disabled. Spending Lane 2/3 on 80–90% of R0/R1 traffic is wasted verification compute *and* alert fatigue. Pricing verification by blast radius is the cost model: the expensive work is reserved for the fraction that can move money or leak an ACL'd span. Shadow mode is the default deployment, so the first artefact a sceptical stakeholder receives is the counterfactual — *would have held N, of which M were true positives* — not a block.

Net of verification vs waste is also not a slide percentage ([QA.md](#) D2). Day one we can name the spend that grounded nothing. Net savings is a Phase-0 output, not an input.

5. Phased roadmap

Enforcement is earned per route, never switched on globally. Shadow is the default ([ARCHITECTURE.md](#) §5.9).

Phase 0 — Shadow (weeks 0–6)

- Deploy the OpenAI-compatible reverse proxy + context-assembly SDK hook on one support route and one acting route.
- Real-emit: gated vs ungated. No user-facing hold.
- Every deterministic mechanism works from the first request: span membership, entitlement, arithmetic, typed interlocks ([QA.md](#) C3).
- Output: the counterfactual *would have held N, of which M were true positives*, plus dead-compute named per trace.
- Capture a human-override baseline (needed later for the 3× auto-rollback rule).
- Exit: platform/SRE accepts p50/p95 on R0/R1; risk accepts the evidence-packet format.

Phase 1 — Enforce R3 (weeks 6–12)

- Turn the matrix on for **R3 only**: payment, deletion, publication, regulated advice.
- Fail closed or escalate when evidence does not arrive in time.
- Co-pending R1 text on the same response still follows the matrix (the refund demo: Edit the customer-visible line, hold the payout).
- Surgical Edit only; no free-form rewrite.
- Exit: zero ungated R3 actions; override rate inside baseline; ledger verify_chain holds.

Phase 2 — Enforce R2 (weeks 12–20)

- Reversible writes and external sends (the copilot “draft email to partner” class).
- Autonomy downgrade (act → propose) instead of blocking ordinary work.
- Per-route circuit breaker: when gate-fail rate exceeds budget, the route drops an autonomy tier until it recovers.
- Plane 2 NLI binding for textual claims on high-R traffic only.
- Exit: R2 Edit/Escalate cells match shadow predictions within the canary window.

Phase 3 — Feedback loops and FNR publishing (from week 16, overlapping)

- Stratified shadow audit: **100% of holds and escalations** plus a random slice of passes, sampled to expensive ground truth.
- Publish the plane’s own **per-route false-negative rate** with confidence intervals — not what we caught, what we missed. Format below; values filled only with measurements.
- Every threshold change: shadow replay over the last N traces showing FP/FN delta; canary by route; **auto-rollback if human-override rate exceeds 3× baseline**.
- Override capture is the feedback loop the brief asks for: humans correct a packet, the correction is a labelled span, calibration moves, the DAG versions.
- Plane 3: semantic-entropy on ungrounded routes; counterfactual bias replay on decision-shaped routes.

bullet geographic / sector policy packs land here as **content** on the versioned DAG, not as a rewrite of the engine.

Gate report — empty schema (the credibility play)

```
route                <id>
window               <start>--<end>
volume               <n>
holds                <n>   escalations <n>   edits <n>   blocks <n>
shadow_would_have_held <n>
true_positives_in_holds <n>   (sampled)
false_negatives       <n>   (sampled passes that should have held)
FNR                   <measured>% ± <CI>   per route
dead_compute_share    <measured>% of steps grounding zero accepted claims
override_rate         <measured>   vs baseline <measured>
p50_added_ms          ≤40 (target)   p95_added_ms ≤200 (target)
policy_version        <id>
```

The emptiness is the claim: we know which fields are knowable at design time and which are not.

Explicitly later (not in this prototype)

Named so they are not pretended: OpenAI-compatible proxy in production form; streaming 1–3B claim extractor; NLI Prosecutor (~300M) for textual claims; Economist yield dashboard; Adjudicator/Red Team; full multi-turn session store; Aho-Corasick PII gazetteer; geographic policy packs. Spec follow-ons: [docs/superpowers/specs/2026-08-23-provenance-recorder-design.md](#) §10.

6. Key risks & mitigations

Mapped to [ARCHITECTURE.md](#) §7–8. Residual risks we will say out loud; rejected approaches we will not re-import.

Risk	Where	Mitigation
False assurance on derived / multi-hop claims. Entailment is weakest where the value is highest. A shallow span that “looks similar” marked SUPPORTED is strictly worse than no control plane , because humans stop checking.	§7 — strongest residual risk	Derived claims are never marked SUPPORTED by entailment. Arithmetic/aggregative → recompute from spans. Neither recomputable nor directly entailed → UNKNOWN . UNKNOWN never collapses into SUPPORTED . Verifiers from a different model family than the generator; deterministic checks carry majority enforcement weight. Publish per-route FNR. Prototype: ClaimKind.DERIVED binds to UNKNOWN (tests/test_binder.py).

Risk	Where	Mitigation
Alert fatigue / over-flag. Over-flagging is how every guardrail gets switched off; under-flagging is liability.	§4 matrix; §8 composite score rejected	Hostile verdict, proportionate action. Identical UNSUPPORTED is Pass+annotate on R1-hedged and Escalate on R3-categorical (Appendix A). User surface is three states, never a 0–100 score. You cannot block, edit or escalate on 87. R0/R1 fail open with annotation so ordinary text is not held hostage. Shadow before enforce.
API-only visibility. No weights, logits, or hidden states. Purely parametric answers have nothing to bind to.	§5 deployment; §8 model-emitted citations rejected; QA.md B1	We work at the I/O layer by design: context-assembly hook + reverse proxy. Binding is undefined without an evidence set, and we do not pretend otherwise. Ungrounded routes are <i>declared</i> ungrounded; blast radius still applies — an ungrounded answer can annotate a draft but cannot authorise a payment. Async semantic-entropy is calibration, never the token stream. <i>We do not claim to verify what we were never given. We claim that what we were never given cannot authorise an action.</i>
The plane is bypassed by load, or disabled by the team.	§8 universal fail-open; QA.md C4–C5	Fail stance belongs to the tier. The plane is a single point of failure only for dangerous actions. Hard gate is on actions, not text; R0/R1 — the majority of volume — pass with annotation. Enforcement earned per route.
The gate is bypassed by malformed input rather than load. A binding that cites a span nobody recorded, an action whose ID does not match any routed claim — the cheapest attack is not defeating the verdict, it is arranging for no verdict to exist.	§2.3; Appendix A.4	Closed, and stated plainly as a strengthening. The gate now fails closed on unresolvable spans (a binding citing an absent span earns UNSUPPORTED, and an ACL we cannot evaluate is recorded as violated, not skipped) and on actions with no routed claims (priced through the frozen UNKNOWN column: R0 Pass, R1 annotate, R2/R3 Escalate). A typo in an action ID can no longer commit the irreversible refund. Locked by 9 regression tests in <code>tests/test_fail_closed.py</code> ; matrix cell count unchanged.

Risk	Where	Mitigation
We enforce a wrong ACL perfectly.	QA.md B4	We do not invent access rights and we do not fix IAM. We stop the model silently bypassing the rights the source already carries, and we log principal × source on every entitlement decision so the over-permissioned index becomes visible. Entitlement is always computed by the plane : a caller may supply a finding that raises a verdict, never one that clears a violation the plane computed itself.
Prompt injection makes a false claim “bind.”	QA.md B5	Binding is computed by us, not asserted by the model. Injection can change what the model says; it cannot change which spans were captured, nor the entailment verdict, nor the ACL. The attack that does work is poisoning a source document — a supply-chain attack on the corpus, forensically traceable via source ID and content hash. <i>We defend the claim-to-evidence link, not the truth of the evidence.</i>
Speculative release with post-hoc recall.	§8	Rejected. Hold-back buffer (~150–300 ms trailing delay). Failures inside the buffer never reach the user. No liability gap.
Integration cost discovered after a “drop-in” sale.	QA.md C2; NARRATIVE.md §5	Said out loud: one SDK hook plus a proxy; days, not quarters, on a standard retrieval stack; not zero. The integration cost is the moat.

Nine mechanisms were removed outright ([ARCHITECTURE.md](#) §8). We will not re-import them under Round 2 pressure: model-emitted citations, speculative release, universal fail-open, cosine-similarity as a verdict, rolling z-score cost anomaly, GBDT rework predictors, demographic parity vs undefined base rates, “interrogator” LLMs, composite risk scores — plus LLM-as-judge on the critical path, confidence/logprob as the hallucination signal, static keyword guardrails as the product, dashboards as the deliverable, debate/voting for real-time gating, uniform validation depth, and any design that assumes access to weights, logits or fine-tuning.

7. Explicit assumptions

Stated so they can be attacked directly.

1. **API-consumed foundation models.** The enterprise does not own the weights. ControlPlane sits at the input/output layer: context-assembly hook + OpenAI-compatible proxy. No logits, no hidden states, no fine-tuning. This is a constraint we designed for, not a gap we paper over ([QA.md](#) B1, C2).

2. **Mixed data governance.** A minority of sources are systems of record with real ACLs (orders DB, vendor agreement). A majority are loosely governed (FAQ, wiki, CRM notes, pasted tickets). The plane must still freeze whatever was actually retrieved — including the loose sources — because those are the spans the model was

allowed to know. Entitlement is only as good as the ACL the source carries; see §6.

3. **India / enterprise regulated workflows as the primary lens.** Running example is an INR refund on a vendor-agreement clause; irreversible actions are payment, deletion, publication, regulated advice. Data-protection frame: Digital Personal Data Protection Act, 2023, plus sector rules on the acting routes (payments). We do **not** ship a hardcoded India/EU/US rule catalogue in the prototype — regulations differ by geography and age quickly, which is why policy is a **versioned DAG of 4-tuples** (`signal`, `threshold`, `action`, `latency_budget`), not nested conditionals ([ARCHITECTURE.md §4](#)). Geographic packs are Phase 3 content.

4. **Tens of thousands of interactions per week** across three concurrent use cases (support, copilot, decision-support), with 80–90% at R0/R1. Directional, from the Round 2 brief, not a measured production trace.

5. **Caller identity is available at request time.** Entitlement is set-membership against `principal.clearance`. If the host application cannot name the principal, the entitlement axis degrades and we will say so rather than fake an ACL.

6. **The prototype is fixtures-and-Lane-1.** Claims in the demos are authored, not extracted by a 1–3B model. Binding is exact/fixture lookup, not NLI. That is sufficient to prove the keystone and the matrix; it is not a production claim extractor.

7. **A missing decision is treated as a refusal, not a permission.** We assume no upstream component is trustworthy enough to earn a pass by omission. If a binding cites a span the ledger never recorded, the claim is `UNSUPPORTED` and its ACL is recorded as violated rather than skipped; if an action carries no routed claims, it is priced through the frozen `UNKNOWN` column rather than passing. This is stated as an assumption because it has a cost we accept: a genuinely harmless action that our fixtures fail to route will be annotated or escalated rather than waved through. We would rather explain a held action than a committed one. Locked by `tests/test_fail_closed.py` (Appendix A.4).

8. Round 2 “Real-World Complexity” coverage

Every Track 1 bullet from the Round 2 brief, addressed or explicitly deferred.

Brief complexity	Stance	Where
Different use cases have different risk tolerance and latency budgets; one-size-fits-all fails	Addressed. Blast-radius matrix + budget-follows-R. Prototype: three use cases, three actuators.	§2.3, §4, Appendix A
Bias, hallucination and privacy overlap in practice	Addressed. Same graph, different labels; fabricated PII is <code>UNSUPPORTED</code> <i>and</i> a no-span leak; true HR fact is <code>SUPPORTED</code> <i>and</i> an ACL miss. Bias kept, as route-level counterfactual flip rate with CI, <code>async</code> — never a per-response verdict.	§2.5
No reliable real-time ground truth	Addressed. Default <code>UNSUPPORTED</code> ; absence of evidence (clause 7.2 does not exist) stays <code>UNSUPPORTED</code> , not <code>CONTRADICTED</code> . The plane does not need a gold answer to refuse an unproven action.	§1, §2.1, Appendix A

Brief complexity	Stance	Where
Over-flagging vs under-flagging	Addressed. Hostile verdict, proportionate action. Encoded in the matrix, not “solved away.” Three user states, no scores.	§2.3, §6
Multi-turn conversations and agents that take actions; compounding risk	Addressed in mechanism; session store deferred. Each pending action on a response is priced separately (refund: R1 Edit + R3 Escalate). An unproven claim cannot authorise the <i>next</i> tool call on that ledger. Deferred: a full multi-turn session store that parents ledgers across turns. Reason: the primitive is per-request frozen provenance; wrapping it in a chatbot memory layer without that freeze would be slideware. Roadmap after Phase 2.	§2.3, §5, Appendix A
Regulatory expectations differ by geography and industry and evolve	Addressed in mechanism; packs deferred. Versioned policy DAG, not hard-coded rules. Shadow replay + auto-rollback on threshold changes. Deferred: geographic/sector policy <i>content</i> packs (DPDP, EU AI Act, sector payment rules as data). Reason: that is a catalogue on top of the engine; shipping a fake pack would age in the room. Primary lens is India/enterprise (assumption 3).	§5 Phase 3, §7.3
Foundation model consumed via API; checker limited to the I/O layer	Addressed. Keystone is the context-assembly hook. No weights, logits, or fine-tuning. Parametric-only routes are declared ungrounded; they cannot authorise R2/R3.	§2.2, §2.4, §6, §7.1

Solutioning areas from the brief, mapped without expanding scope:

Brief area	Our answer
Detection	Binding against provenance (not LLM-as-judge). Deterministic recompute for numeric/structural. Derived → UNKNOWN. Leakage = set-membership. Bias = counterfactual invariance async. Rejected on the critical path: AI-as-judge, confidence scores, cosine thresholds.
Decision logic	Frozen $R \times S$ matrix. Actuators Block / Edit / Escalate / Pass. Pure rule engine at decision time — zero LLM reasoning.
Architecture	Inline middleware on the commit path; text streams behind a hold-back; three lanes; speculative <i>verification</i> never speculative <i>release</i> .

Brief area	Our answer
Governance	Versioned DAG, policy_version on every ledger, hash-chained audit trail, canary + 3x-override rollback.
Feedback loops	Override capture → labelled spans → shadow replay → versioned threshold. Phase 3, not coded in the MVP.
Metrics	Publish FNR per route , dead-compute share, override rate, p50/p95. Empty schema until measured.

Appendix A — Prototype evidence

Working proof of the keystone on illustrative fixtures. Not production-grade, not real enterprise data, and not a claim extractor. Commands below were run for this proposal; actuators are locked by tests.

A.1 Dual-action refund (Edit + Escalate)

```
python3 examples/refund_trace_demo.py
```

Expected actuators (do not invert):

Pending action	Tier	Driving finding	Matrix cell	Actuator
show_text — Show text to the customer	R1	hr_side grounded on span ACL {hr-confidential} while caller clearance is {vendor-public}	R1 × Contradicted entitlement violation	Edit — unentitled span stripped
issue_refund — Issue the refund (irreversible, █1,84,000, ORD-9)	R3	clause_72 (“Clause 7.2 permits this refund”) binds to no span	R3 × Unsupported + categorical	Escalate — held and escalated with the evidence packet

Never say the refund was “blocked.” R3 × unsupported-categorical = Escalate.

Observed in the demo run:

- bullet Delegated response: Approved. Refund of █1,84,000 issued under clause 7.2 of the vendor agreement.
- bullet Principal cs-agent-17, clearance {vendor-public}, policy matrix-v1.
- bullet Five spans at context assembly; **no span for clause 7.2**.
- bullet clause_72 verdict **UNSUPPORTED** (absence of evidence, not conflicting evidence — the clause does not exist, it does not “cap” or “deny”).
- bullet hr_side verdict SUPPORTED via span-3 **and** entitlement **VIOLATION**.
- bullet hrq_search / crm_lookup produce spans that ground no accepted claim (dead compute, named by walking the graph backward).
- bullet show_text actuator **Edit**; driving hr_side, clause_72; evidence packet attached.
- bullet issue_refund actuator **Escalate**; driving clause_72; packet carries the unproven clause.
- bullet verify_chain() = True.

Locked by tests/test_refund_scenario.py (show_text → Edit, issue_refund → Escalate, chain holds).

A.2 Three use cases, three R-tiers, three actuators

Round 2 reference: customer support assistant, internal knowledge assistant, decision-support tool.

```
python3 examples/multi_usecase_demo.py
```

Expected actuators:

Use case	Action	Tier	Claim	Cell	Actuator
Customer support chatbot	show_reply	R1	hedged warranty guess, UNSUPPORTED	R1 × Unsupported + hedged	Pass + annotate
Internal knowledge copilot	draft_partner_email (external send)	R2	categorical partner SLA, UNSUPPORTED	R2 × Unsupported + categorical	Edit
Decision-support refund	issue_refund	R3	“Clause 7.2 permits this refund,” UNSUPPORTED	R3 × Unsupported + categorical	Escalate

Observed in the demo run:

1. Customer support chatbot
action show_reply (Show reply to the customer, R1)
cell R1 × Unsupported + hedged → Pass + annotate
actuator Pass + annotate
2. Internal knowledge copilot
action draft_partner_email (Draft email to partner (external send), R2)
cell R2 × Unsupported + categorical → Edit
actuator Edit
3. Decision-support refund
action issue_refund (Issue the refund, R3)
cell R3 × Unsupported + categorical → Escalate
actuator Escalate

Same plane. Three R-tiers. Three actuators. Hash chain verifies on each ledger.

Locked by tests/test_multi_usecase.py.

A.3 What the prototype is — and is not

In this slice	Not in this slice
Provenance Recorder (record step/span, freeze context)	OpenAI-compatible reverse proxy
Append-only hash-chained Evidence Ledger	Streaming 1–3B claim extractor
Deterministic binder, default UNSUPPORTED; derived → UNKNOWN	NLI Prosecutor / semantic-entropy probe
Entitlement Auditor (ACL ⊂ clearance)	Counterfactual bias replay (specified, async, not coded)
Action Interlock on the frozen matrix	Economist live breaker / FNR publisher

In this slice	Not in this slice
Refund dual-action + three-use-case fixtures	Multi-turn session store, geographic policy packs, PII NER model

No LLM and no network on the critical path. That is a fidelity choice, not a missing detector: Lane 1 is where 80–90% of volume and all of R3 entitlement/interlock already live.

A.4 Fail-closed regressions (three silent-pass paths closed)

```
pytest -q tests/test_fail_closed.py
```

Three paths could previously let an action reach **Pass** without a verdict. All three now fail closed, and the frozen matrix was not widened to do it:

Path	Old behaviour	Now
A binding cites a span the ledger never recorded	earned SUPPORTED on an asserted citation	UNSUPPORTED (fixture-unresolved) — a binding may only cite spans this ledger recorded
The same unresolvable span in the entitlement audit	audit skipped , which reads as “not violated”	recorded as violated (UNRESOLVABLE_SPAN) — an ACL we cannot evaluate is not one we may ignore
An action with no routed claims (e.g. a typo'd action ID)	empty matrix column, actuator Pass	routed through the frozen UNKNOWN column: R0 Pass, R1 annotate, R2/R3 Escalate with an evidence packet

Also closed: a caller-supplied findings map (including an empty one) can no longer disable entitlement. Supplied findings may raise a verdict; they cannot clear one the plane computed.

Observed: 9 regression tests pass; the typo'd R3 refund is **held and escalated** rather than committed; the dual-action refund is unchanged (show_text → Edit, issue_refund → Escalate, both independent); verify_chain() = True; matrix cell count unchanged. Gate latency is unmoved — see submission/latency_bench.json (p50 **0.074 ms**, p95 **0.134 ms** gate-internal, against a **≤40 ms p50 / ≤200 ms p95** budget; never quote 40 ms as p95).

This is a strengthening, not a workaround: the plane's claim is that an unproven action cannot commit, and these were the three ways an action could avoid being proven at all.

Design is closed in [ARCHITECTURE.md](#). This proposal is a rendering of that file for Round 2, plus evidence that the keystone runs.

Appendix B — Live control plane (production elevation)

The core mechanism is no longer CLI-only. On branch main:

```
uvicorn controlplane.server.app:create_app --factory --host 127.0.0.1 --port 8787
# Judge console: http://127.0.0.1:8787
# Or: docker compose up --build → http://localhost:8080
```

Surface	Proof
OpenAI-compatible gate	POST /v1/chat/completions with scenario=refund returns Edit+Escalate in controlplane extension
Enforce overlay	User-visible text strips clause 7.2; irreversible refund action_allowed=false
Shadow / FNR	GET /v1/controlplane/metrics publishes would-have counters and labeled FNR shape
Policy packs	policies/*.yaml — support / copilot / decision-support
Audit	GET /v1/controlplane/requests/{id}/audit.jsonl

Lane 1 remains deterministic. No LLM on the critical path. Matrix transcribed, never redrawn.